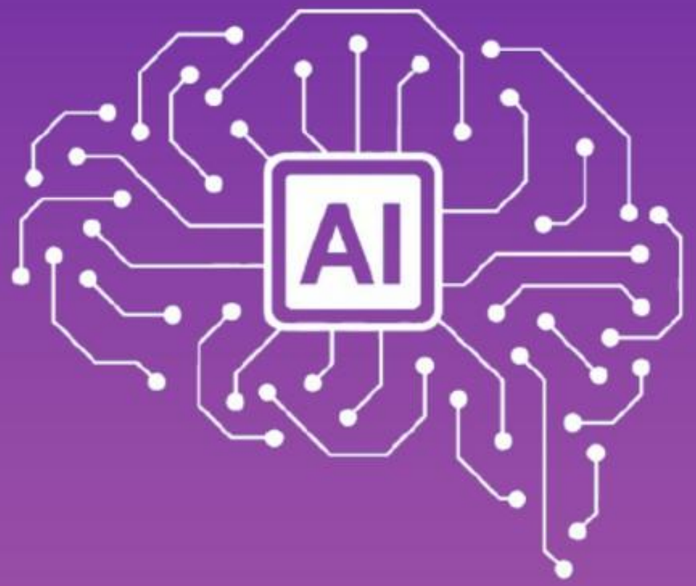
The Driving Factors for Integrating AI in the Sustainable Development of Indian SMEs



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Introduction

- SMEs are the economic backbone of India.
- AI adoption is growing in business but underutilized in SMEs.
- Sustainability is a key challenge: emissions, waste, energy.

Objective of the Project

1. To determine whether AI-enabled machines reduce emissions, waste and energy use without affecting output.

2. To identify key sustainability factors that influence AI adoption in SMEs.

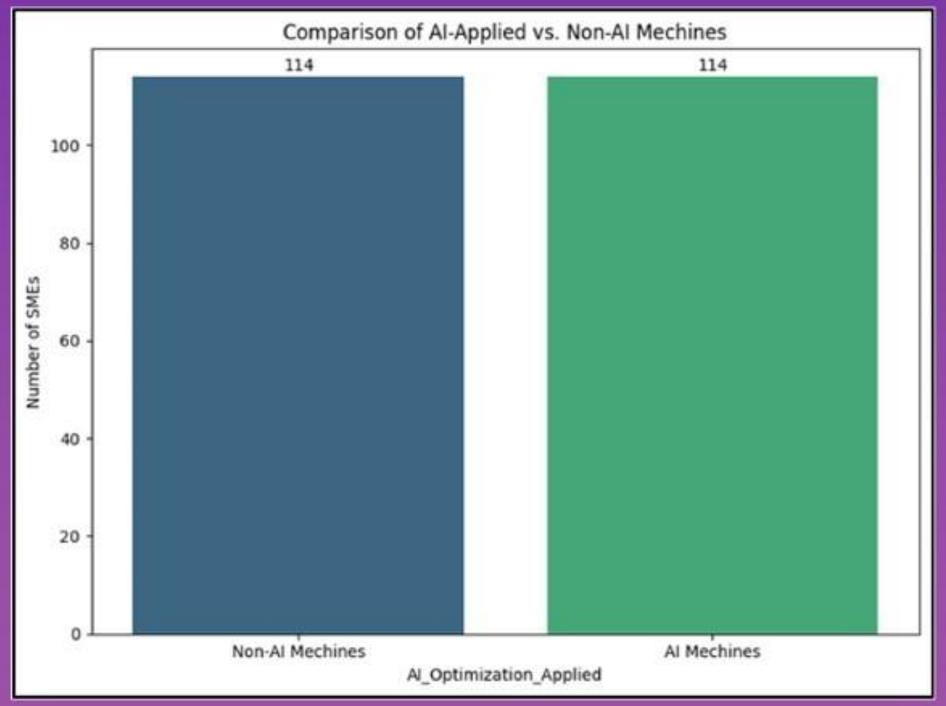
Literature Review

1. BDA helps businesses make evidence-based decisions and optimize resources (Maran, 2023)
2. AI+ML+IoT in Industry 4.0 can enhance SME growth despite skill and cost barriers (Bhavana Raj, 2023)
3. Lack of standardization and accessibility limits AI use in SMEs (Silva, 2022)
4. Low-code platforms help SMEs adopt AI without needing technical expertise (Kok, 2024)
5. Edge AI helps SMEs in real-time decisions and global competitiveness (Mallela, 2024)
6. ML models like Random Forest improve forecasting in e-commerce (Biswas, 2025)
7. SMEs lag in AI adoption in early product development due to lack of trust/support (Cooper, 2024)
8. Ethical AI tools like XAI ensure risk evaluation and fairness in SMEs (Han, 2023)
9. AI-driven geospatial tools aid SMEs in expansion and market targeting (Mallela, 2024)
10. AI helps monitor UN SDGs using accurate and scalable tools (Miller, 2020)

Literature Gap

1. Most studies focus on reviews or theoretical models
2. Most data used is from large firms, not SMEs
3. Few practical studies on AI/ML implementation in real SME environments

Dataset Overview



1. Source: Kaggle (SME Machine Data)
2. Features: Carbon Emissions, Energy Consumption, Waste Generated, etc.
3. 228 entries with mixed categorical & numerical data.

Tools and Technologies

1. Python (Jupyter Notebook, Anaconda)
2. Libraries:

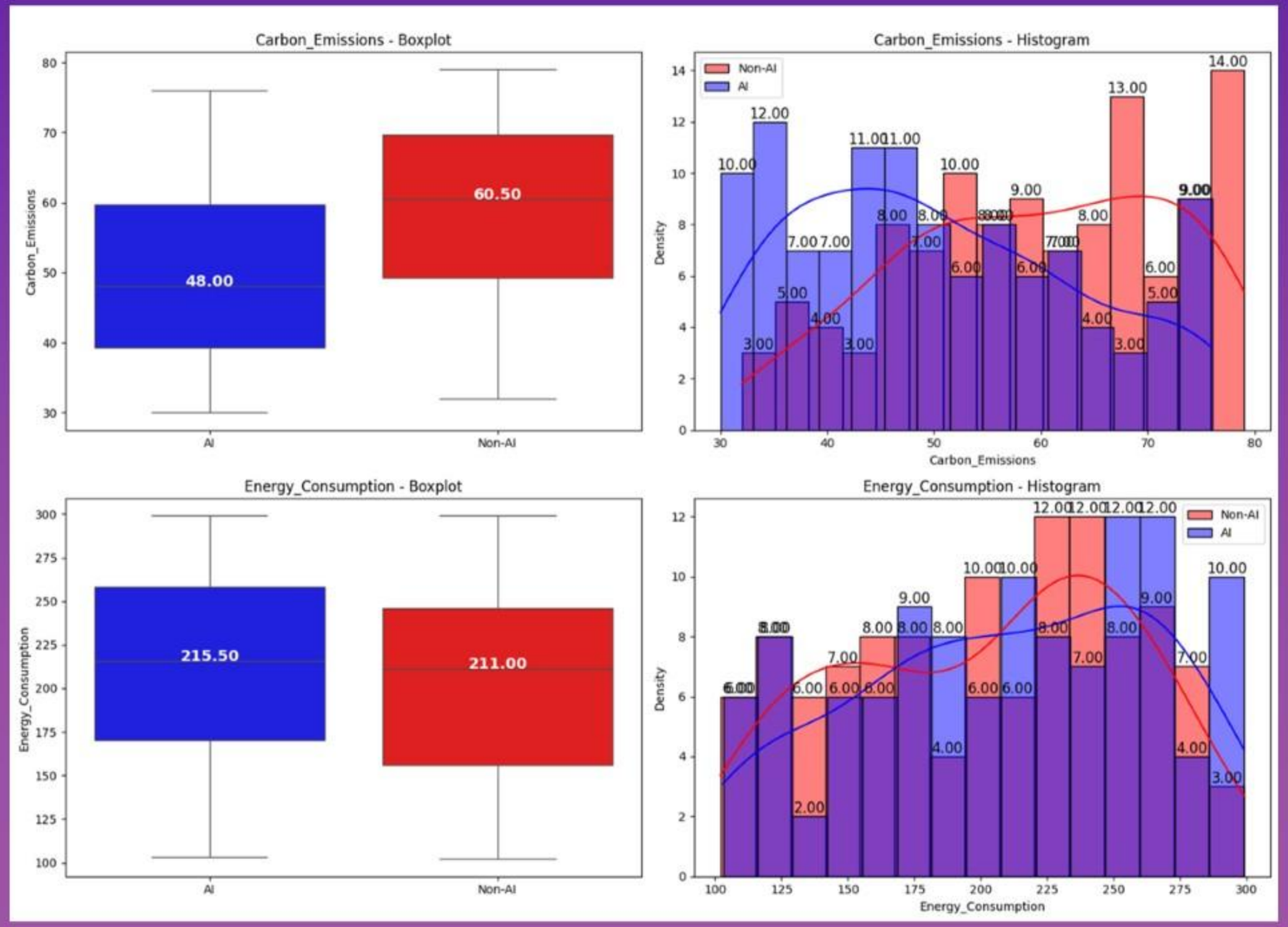
- NumPy & Pandas (data handling)
- Seaborn & Matplotlib (visualization)
- Scikit-learn & SciPy (ML & stats)

Research Methodology

1. Exploratory Data Analysis
2. Statistical Analysis
3. Machine Learning Model:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM, KNN
- Clustering: K-Means

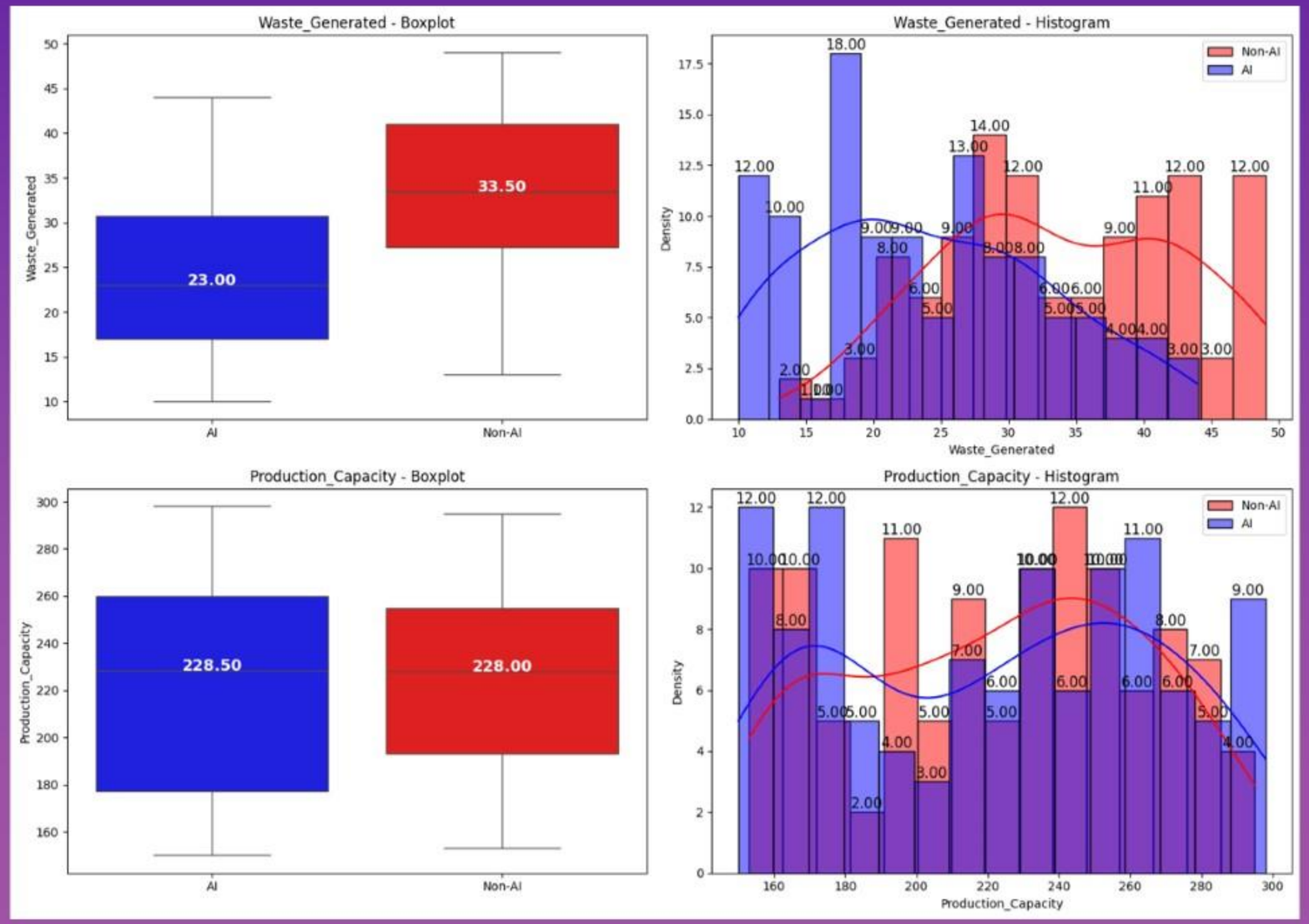
Box plot & Histogram of Sustainability Indicators



Reveals that AI machines:

- **Emit significantly less carbon (48 vs. 60.5).**
- **Use slightly more energy (215 vs. 211).**

Box plot & Histogram of Sustainability Indicators

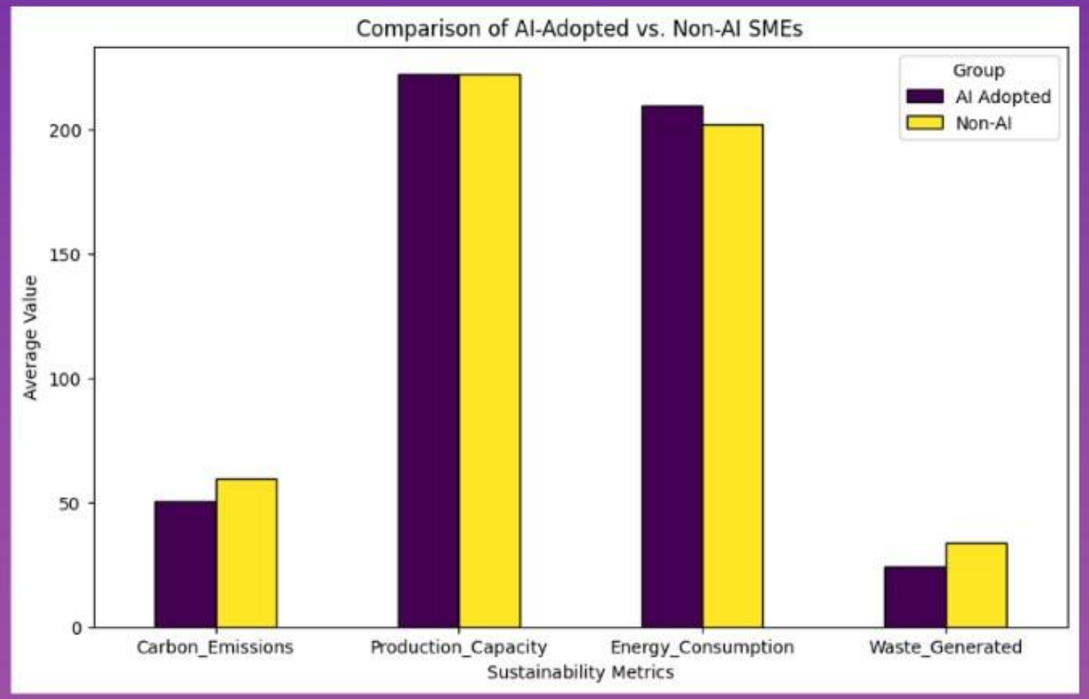


Reveals that AI machines:

i. **Maintain similar production capacity (228.5 vs. 228).**

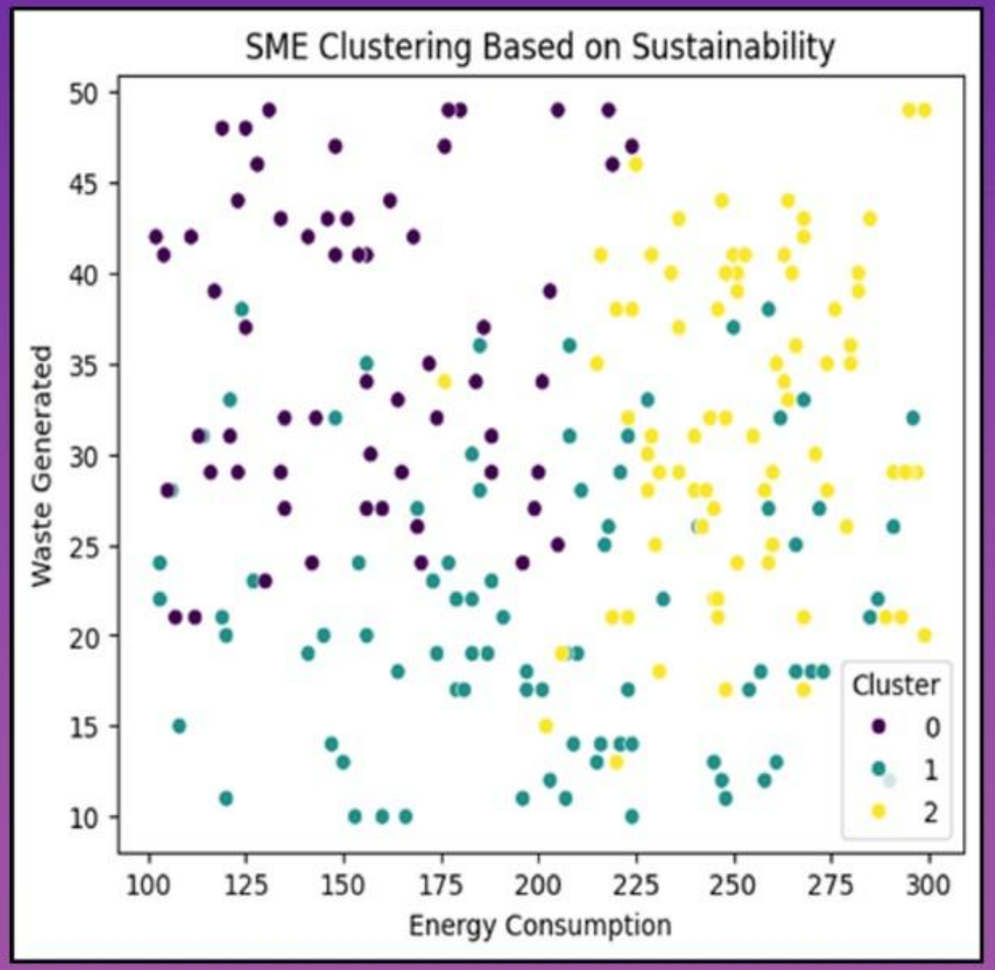
ii. **Generate less waste (23 vs. 33).**

Bar Chart of Sustainability Indicators



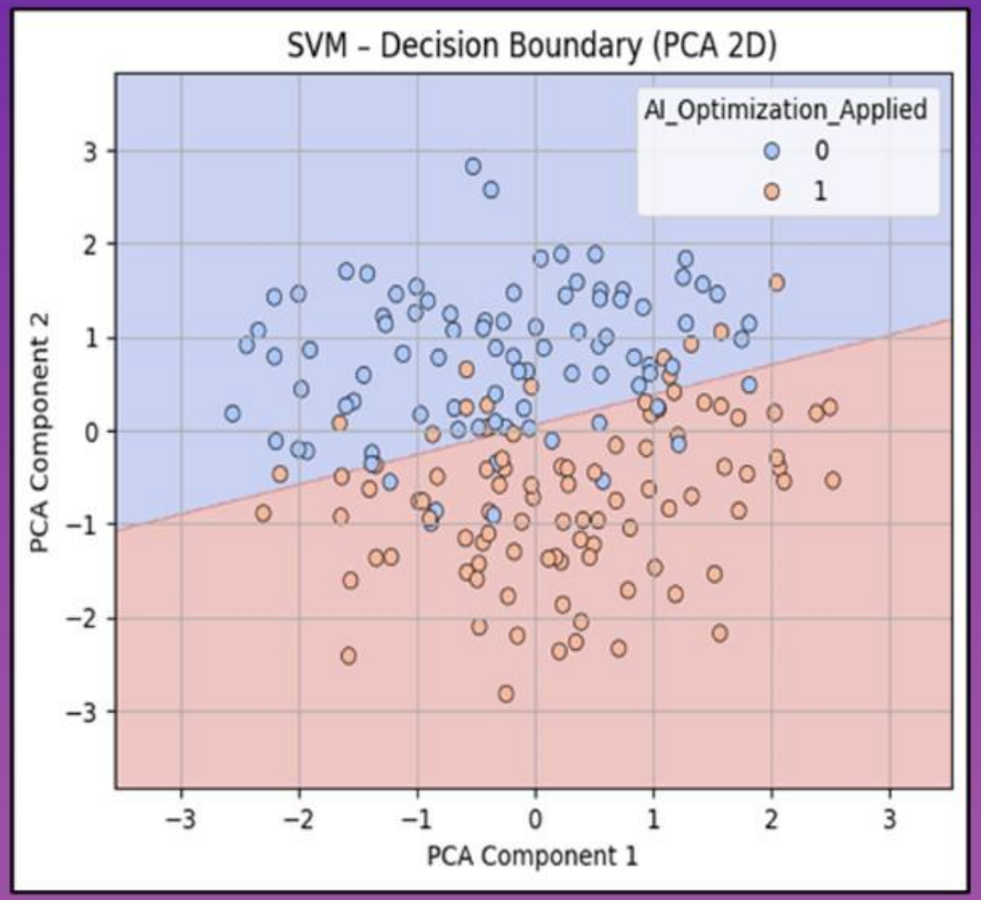
SN	Parameters	AI machine	Non AI machine	Interpretation
01	**Carbon Emissions**	48	60.50	AI machines emit significantly less carbon, indicating improved environmental efficiency.
02	**Production Capacity**	228.5	228	Almost the same, suggesting AI improves sustainability without compromising output.
03	**Energy Consumption**	215	211	Slightly higher in AI machines, possibly due to additional computational tasks.
04	**Waste Generated**	23	33	AI machines produce less waste, improving resource efficiency.

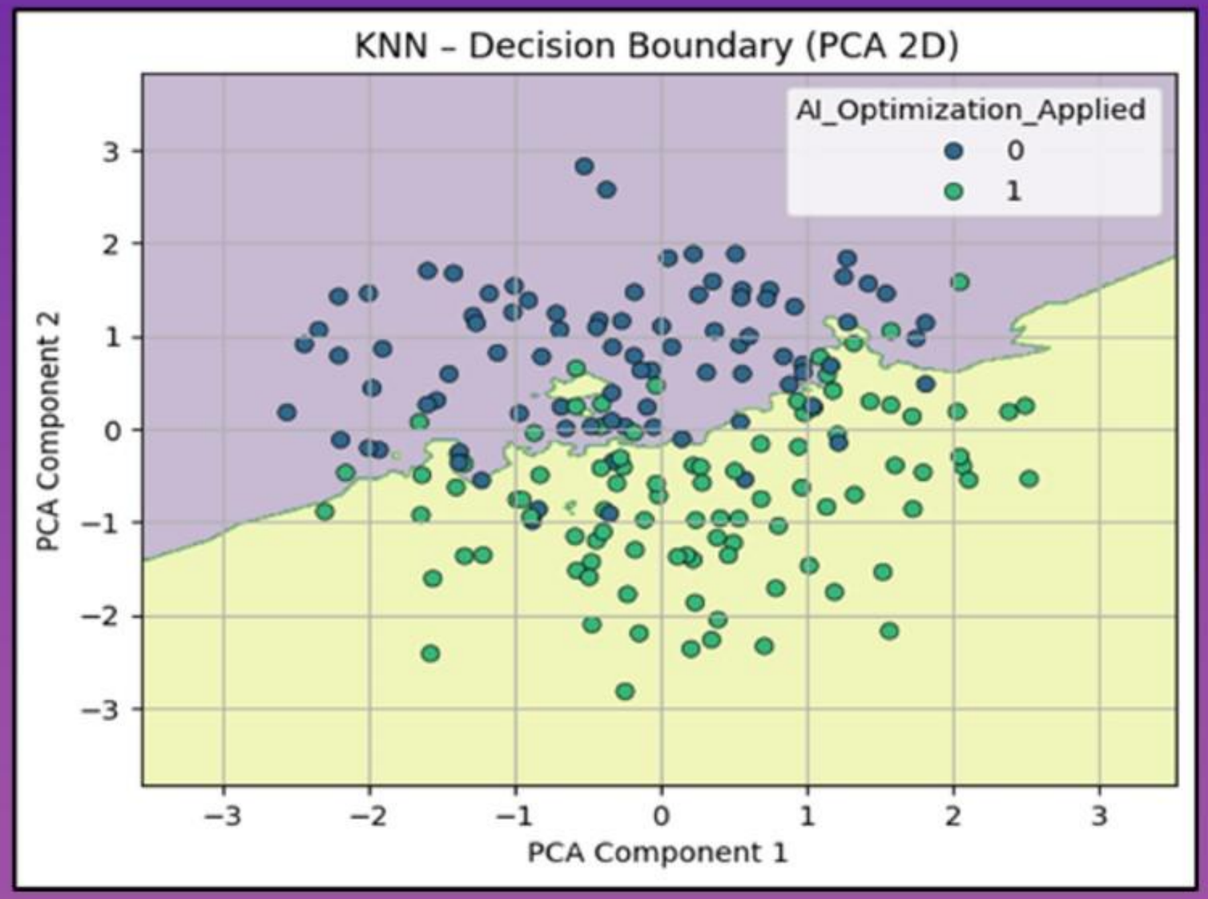
Scatter Plot of Sustainability Indicators



- Cluster 0 : higher waste generation and varied energy usage (Moderate performance).
- Cluster 1 : Low waste and moderate energy use (Most sustainable).
- Cluster 2 : Higher energy consumption but varied waste (Possibly less efficient).

Machine Learning Methods





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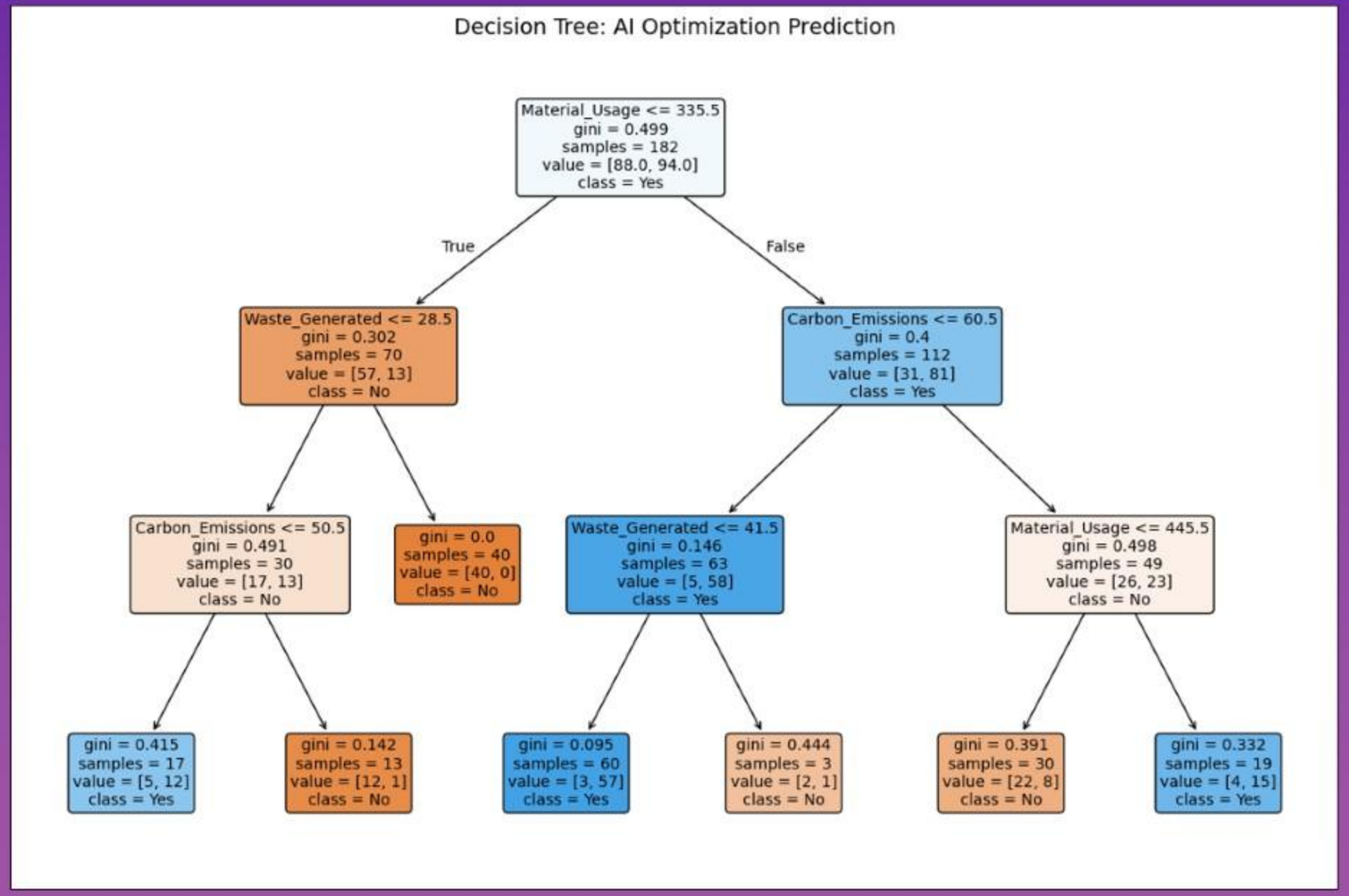

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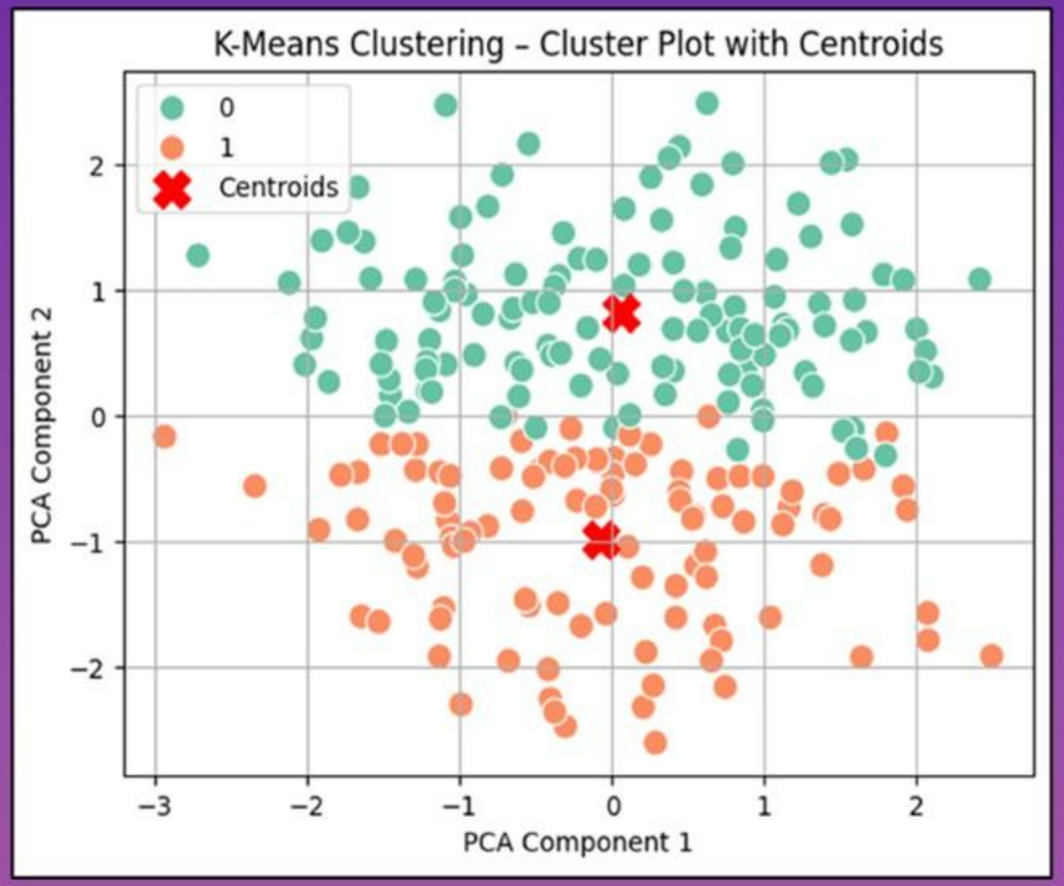

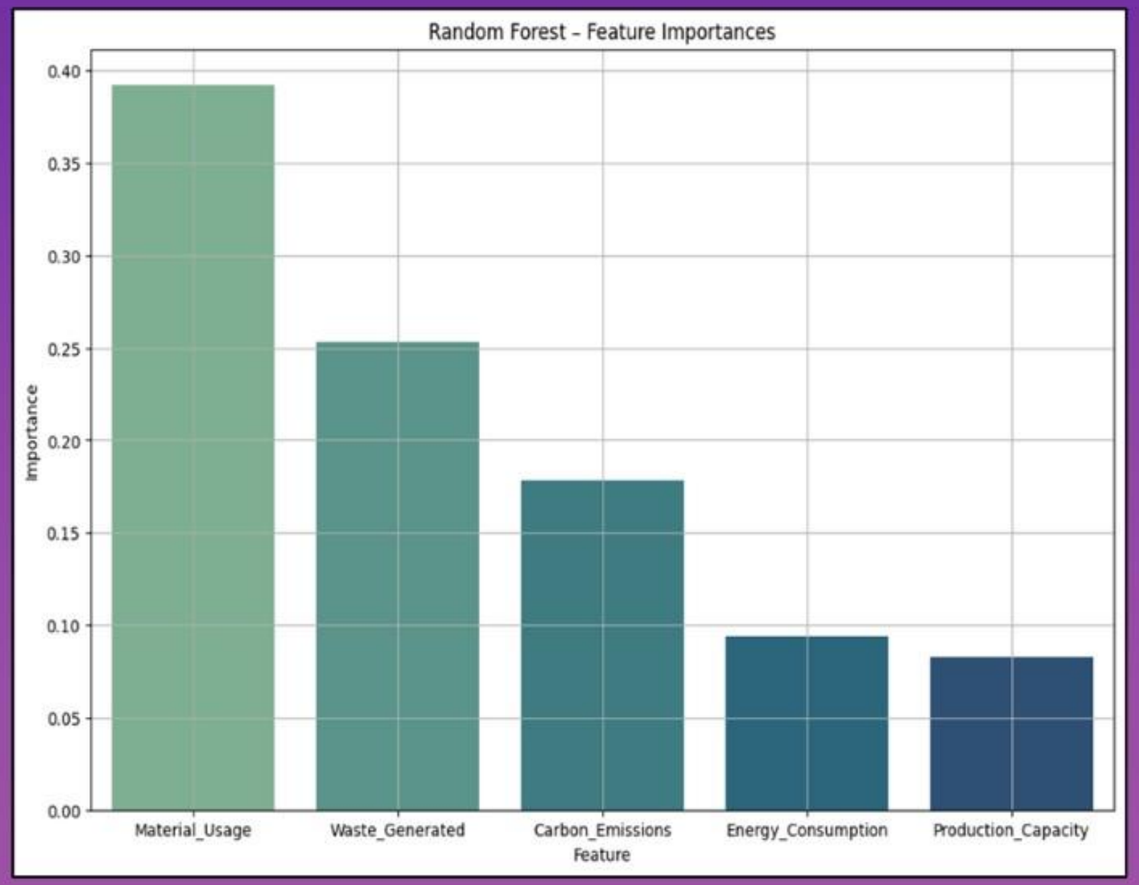

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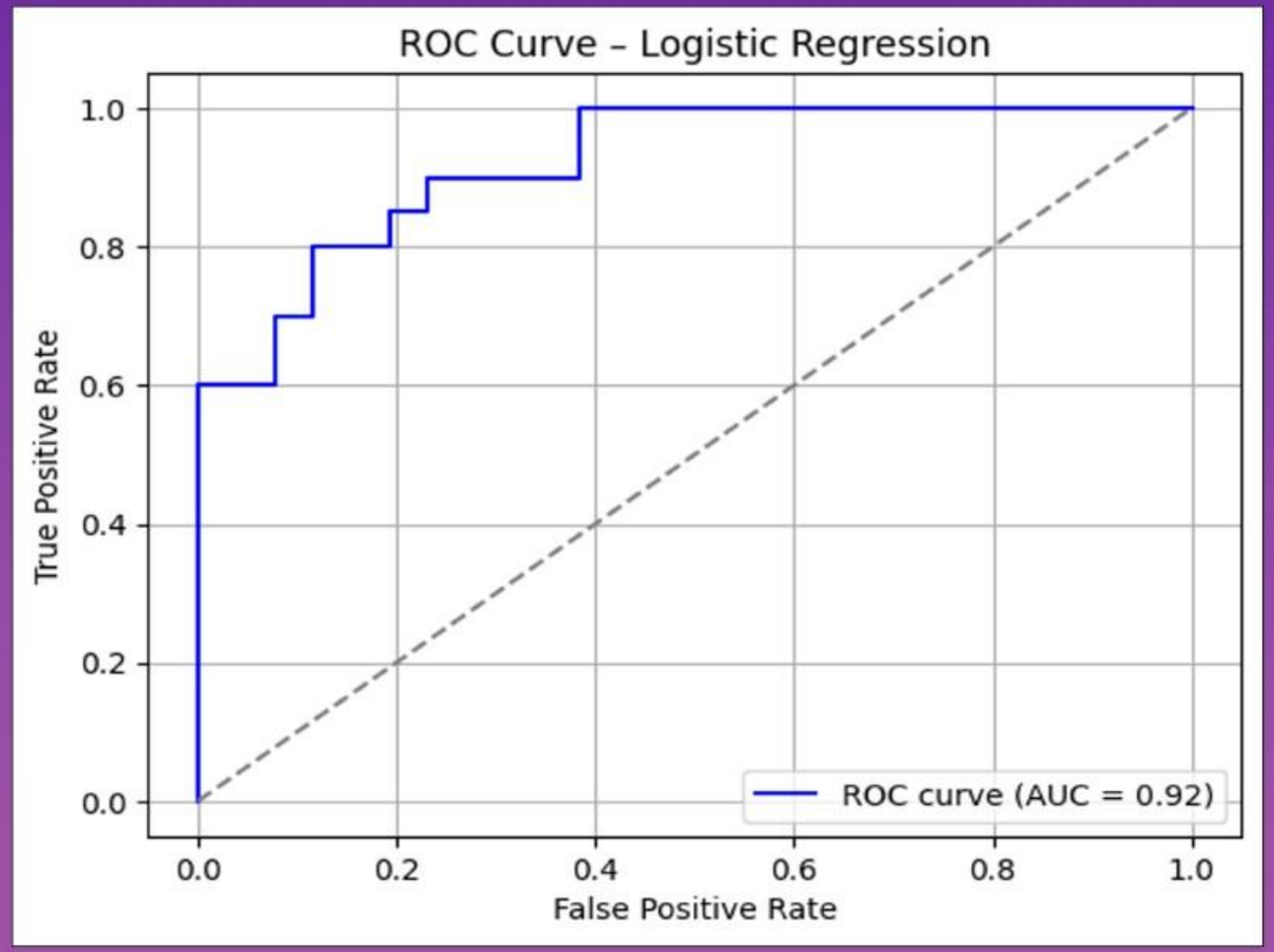
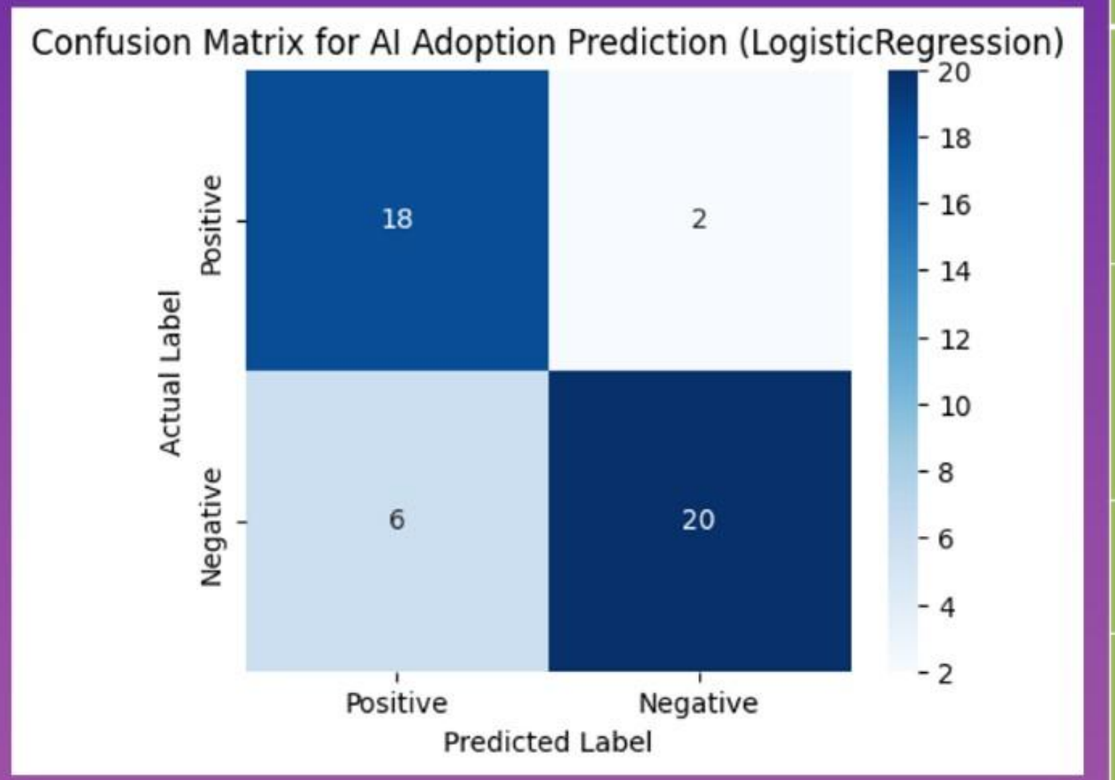


Machine Learning Results

Model	Accuracy	Comments
Logistic Regression	83%	• Best performing overall. • Suitable for in this case.
SVM	72%	• Lower accuracy, precision & recall. • Not suitable for this type of problem.
KNN	70%	• Lowest accuracy. • Not suitable for this type of problem.
Decision Tree	80%	• Balanced performance. • Good if you want decent performance.
Random Forest	78%	• Fair balance accuracy. • Reliable, but less accurate than Decision Tree
K-Means Clustering	59%	• Lower performance overall. • Not preferred in this case.

Confusion Matrix for Logistic Regression



Predictions	Value	Comments
True Negatives	20	The model reliably distinguishes non-AI users, which helps in targeting them for potential AI adoption initiatives.
False Positives	6	The model might have confused some efficient or sustainable SMEs as AI users even though they aren't.
False Negatives	2	A small number of AI adopters were missed.
True Positives	18	The SMEs that are genuinely AI-optimized, and the model was able to recognize them correctly.

Key Findings

- AI machines: lower emissions, less waste ,no drop-in production capacity.
- AI helps businesses produce more sustainably by reducing emissions and waste while boosting efficiency.

Managerial Implications

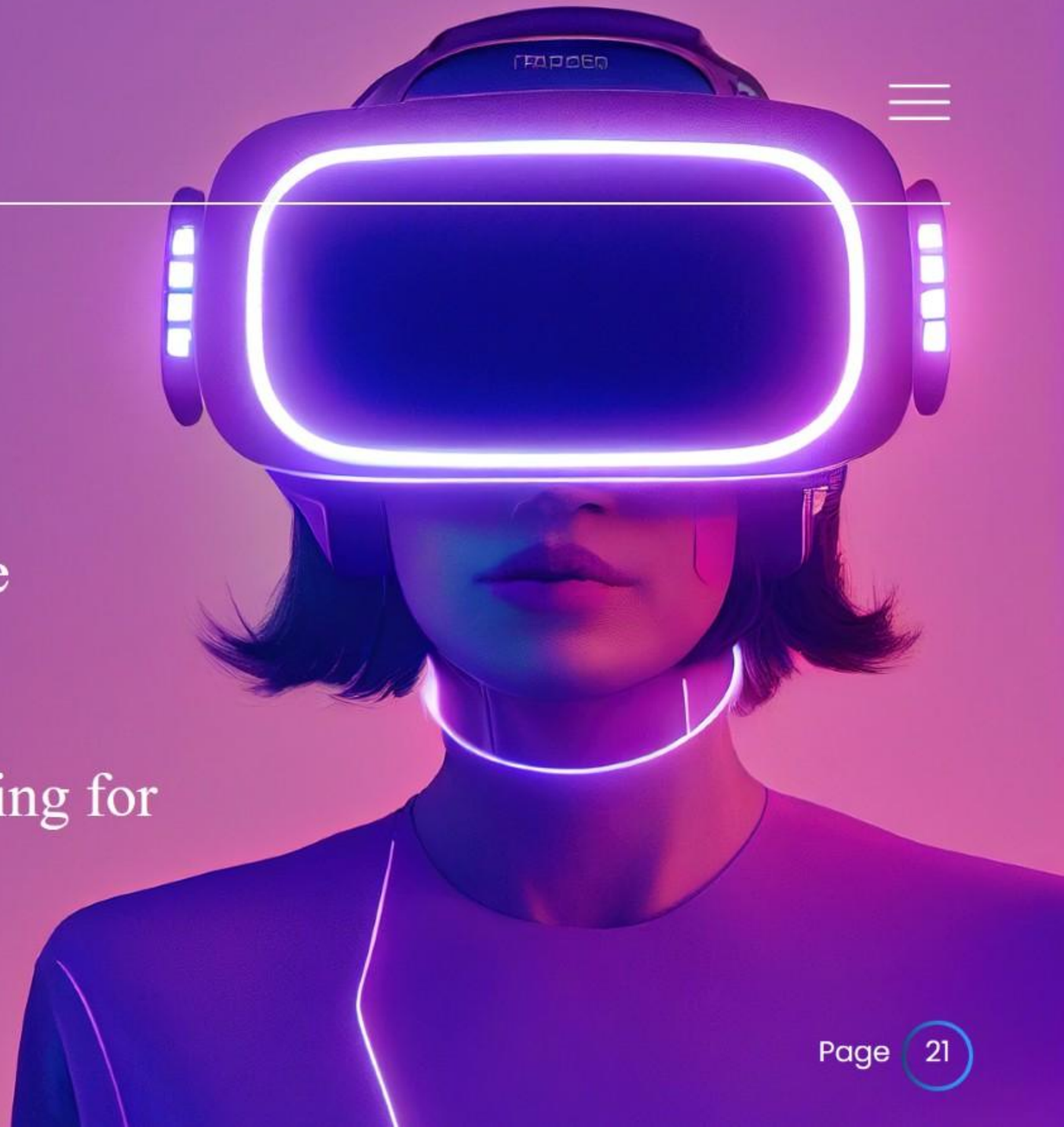
- Managers can use AI to track machine performance
- Data-driven decisions = better resource use
- Scope for policy & funding support for AI in SMEs

Limitations & Future Scope

- Dataset limited to secondary machine data
- Lacks industry-specific variation
- Future: Real-time IoT-AI integration, dashboards (Power BI/Tableau)

Conclusion

1. AI positively influences SME sustainability.
2. Logistic Regression most reliable predictor.
3. Supports informed decision-making for sustainable growth.



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THANK YOU